MIL-P-24138A(SH)
30 June 1986
SUPERSEDING
MIL-P-24138(SHIPS)
11 March 1965
(See 6.5)

MILITARY SPECIFICATION

PASSIVATOR COMPOUND, NAVY BOILER

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a uniform mixture containing essentially dust free, granular sodium nitrite, monosodium phosphate and disodium phosphate. No other materials shall be incorporated into the passivator.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and the supplement thereto, cited in the solicitation.

SPECIFICATION

FEDERAL
PPP-D-723 - Drums, Fiber.

STANDARDS

FEDERAL
FED-STD-313 - Material Safety Data Sheets Preparation and the Submission of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads.

2.1.2 Other Government documents. The following other Government documents forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be those in effect on the date of the solicitation.

DEPARTMENT OF LABOR
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)
Code of Federal Regulations (CFR)
29 CFR, Part 1910.1200 - Hazard Communication Standard.

DEPARTMENT OF TRANSPORTATION (DOT)
Code of Federal Regulations (CFR)
49 CFR, Parts 170-179 - Hazardous Material Regulations.

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
S9086-T8-STM-000 - Naval Ships' Technical Manual, Chapter 593, Pollution Control.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN CHEMICAL SOCIETY (ACS)
Specifications, Reagent Chemicals.

(Application for copies should be addressed to the American Chemical Society, 1155 16th Street NW, Washington, DC 20036.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Composition. The nominal composition of the passivator is 50 percent sodium nitrite ($NaNO_2$), 25 percent monosodium phosphate ($NaH_2PO_4.H_2O$) and 25 percent disodium phosphate (Na_2HPO_4).

3.1.1 Sodium nitrite. The sodium nitrite used in the passivator shall have an assay of not less than 97 percent. The passivator shall contain not less than 44.0 percent nor more than 54.0 percent sodium nitrite when tested as specified in 4.3.1.

3.1.2 pH. The pH shall be 6.6 ± 0.3 when tested as specified in 4.3.2.

3.1.3 Total phosphoric anhydride (P_2O_5). The total phosphoric anhydride (P_2O_5) shall be not less than 24.7 percent nor more than 25.7 percent when tested as specified in 4.3.3.

3.1.4 Insoluble matter. The insoluble matter shall be not more than 0.25 percent when tested as specified in 4.3.4.

3.2 Material safety data sheet. The contracting activity shall be provided a material safety data sheet (MSDS) at the time of contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313 and 29 CFR 1910.1200, Hazard Communication. When FED-STD-313 is at variance with the CFR, 29 CFR 1910.1200 shall take precedence, modify and supplement FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification (see 6.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the

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contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality conformance inspection.

4.2.1 Lot. A lot shall consist of not more than 20,000 pounds of passivator, or 1 day's production, whichever is smaller.

4.2.2 Examination of filled containers. A random sample of filled containers, taken from each lot in accordance with MIL-STD-105 at inspection level I and acceptable quality level of 2.5 percent defective, shall be examined to verify compliance with all stipulations of this specification regarding fill, closure, marking and any other requirements not involving tests. Samples shall be examined for defects of the container and closure, for evidence of leakage and for unsatisfactory markings. Each sample-filled container shall be weighed to determine the amount of contents. Any container in the sample having one or more defects or under required fill shall not be offered for delivery. If the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, this shall be cause for rejection of the lot represented by the sample.

4.2.3 Sampling procedure for tests. Two filled containers shall be randomly selected from each lot and a 1-pound sample shall be taken from each container. The samples shall not be mixed. Samples shall be marked with the lot number and date of sampling, and shall be subjected individually to the tests specified in 4.3. In addition, representative samples of the ingredients used in fulfilling each contract or order shall be taken as follows: 1-pound each of sodium nitrite ($NaNO_2$), monosodium phosphate ($NaH_2PO_4.H_2O$) and disodium phosphate (Na_2HPO_4) placed in individual containers and marked with a lot number and date sampled. These ingredients shall be used in the event that additional testing for confirmation of composition or quality becomes necessary.

4.2.4 Tests. The two samples from each lot selected (see 4.2.3) shall be subjected individually to the tests specified. Analytical reagent grade chemicals and reagent grade water shall be used throughout the tests. Where applicable, the reagents shall meet the requirements of the American Chemical Society specifications.

4.3 Test procedures.

4.3.1 Sodium nitrite. Sodium nitrite shall be tested as follows.

4.3.1.1 Reagents.

(a) Sulfamic acid, purified - The sulfamic acid shall be purified as follows: Dissolve 125 grams of sulfamic acid in 300 grams of water previously heated to 70 degrees Celsius (°C). Filter three times. With consequent lowering of the temperature, discard the material which crystallizes from the solution. Cool the filtrate rapidly in an ice-salt mixture and let stand for 20 minutes. Remove the crystals thus formed by suction filtration. Wash the crystals with a small portion of ice

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water, then twice with cold ethanol, and finally with ether. Air dry for 1 hour, grind in an agate mortar and store in a desiccator.

(b) Sulfuric acid - 50 percent by volume.
(c) Potassium bromide - Saturated solution.

4.3.1.2 Apparatus.

(a) pH meter, laboratory type with millivolt and polarizing current capability.
(b) Silver billet electrode.
(c) Platinum electrode.
(d) Buret, 50-milliliter (mL) capacity graduated in 0.1-mL divisions, with delivery error no greater than plus or minus 0.05 mL.
(e) Volumetric flask, 1000-mL capacity.
(f) Magnetic stirrer.
(g) Magnetic stirring bar.
(h) Beaker, 400-mL capacity.
(i) Ice-water bath.

4.3.1.3 Procedure. Dissolve to a 20-gram sample of the passivator (weighed to the nearest milligram) in 700 to 800 mL of distilled water (pH of 6.6 to 7.2) in a 1000-mL volumetric flask. Dilute to the mark and mix. Accurately weigh a 0.500-gram portion of the purified sulfamic acid. Transfer the sulfamic acid to a 400-mL beaker and dissolve in 100 mL of cool water (25°C or less). Add 15 mL of the potassium bromide solution. Cool to 15°C in an ice water bath. Cool the sulfuric acid solution in the ice water bath. Connect the silver billet electrode with a polarizing jumper to the input jack of the pH meter. Connect the platinum electrode to the reference jack. Set the pH meter to read millivolts. Connect meter to power supply. Add 25 mL of the sulfuric acid solution to the sulfamic acid solution. Agitate the acid solution by means of the magnetic stirrer and stirring bar. Immerse the active surfaces of the electrodes in the acid solution. Titrate the sulfamic acid solution with the passivator sample solution, with the buret delivery tip just below the surface of the sulfamic acid solution. The equivalence point shall be taken as the point at which further additions of the titrant (passivator sample solution) will not cause any additional surge in the millivolt reading. Calculate as follows:

$$\text{Percent sodium nitrite} = 71130 \frac{W_2}{W_1 V}$$

Where:
W_1 = Weight in grams of passivator sample.
W_2 = Weight in grams of sulfamic acid.
V = Volume of titrant (passivator sample solution), mL.

4.3.2 pH. pH shall be tested as follows.

4.3.2.1 Reagents.

(a) Distilled water, pH of 6.6 to 7.2.
(b) Standard buffer solution, pH of 7.0 at 25°C.

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4.3.2.2 Apparatus.

(a) pH meter, laboratory type.
(b) Glass electrode.
(c) Calomel electrode.
(d) Pipet, 25-mL delivery.
(e) Volumetric flask, 200-mL capacity.
(f) Thermometer.
(g) Beaker, 150-mL capacity.

4.3.2.3 Procedure. Transfer a 25-mL aliquot of the passivator sample solution prepared as specified in 4.3.1.3 to a 200-mL volumetric flask. Dilute to the mark with distilled water and mix. Transfer approximately 100 mL of the solution to a 150-mL beaker. Standardize the pH meter with the pH 7.0 standard buffer solution. Measure the pH of the diluted sample solution at 25°C with the pH meter.

4.3.3 Total phosphoric anyhydride (P_2O_5).

4.3.3.1 Reagents.

(a) Magnesia mixture. Dissolve 55 grams of $MgCl_2.6H_2O$ and 110 grams of NH_4Cl in 500 mL of water. Make distinctly ammoniacal with NH_4OH and let stand overnight. Filter off any precipitate. Acidify with HCl and dilute to 1 liter in a glass-stoppered bottle.
(b) Ammonium hydroxide (1:1), 50 percent by volume.
(c) Ammonium hydroxide (1:19), 5 percent by volume.
(d) Hydrochloric acid (1:1), 50 percent by volume.
(e) Hydrochloric acid (1:19), 5 percent by volume.

4.3.3.2 Apparatus.

(a) Pipet, 25-mL delivery.
(b) Beaker, 400-mL capacity.
(c) Gooch crucible, porcelain, Coors, No. 3, or equal.

4.3.3.3 Procedure. Transfer a 25-mL aliquot of the passivator sample solution as prepared in 4.3.1.3 to a 400-mL beaker. Acidify with hydrochloric acid (1:1). Cool and add 40 mL of magnesia mixture. Add ammonium hydroxide (1:1) dropwise with constant stirring until precipitation starts. Let stand 15 minutes. Add 10 mL of ammonium hydroxide (1:1) per 100 mL of solution, while stirring gently. Allow to stand overnight. Filter through a No. 42 Whatman filter paper, or equal, without attempting to transfer the precipitate to the paper. Wash the residue by decantation and the filter paper a few times with ammonium hydroxide (1:19). Dissolve the precipitate on the filter paper with 30 mL of hydrochloric acid (1:1), catching the solution in the original beaker. Wash the filter paper thoroughly with hydrochloric acid (1:19). Dilute the solution to 100 mL and add 2 to 4 mL of magnesia mixture. Add ammonium hydroxide (1:1) dropwise, stirring until precipitation starts. Let stand 15 minutes. Add ammonium hydroxide (1:1) to 10 percent of excess. Allow the precipitate to settle. Filter on a pre-ignited and weighed Gooch crucible with a glass fiber filter disc, employing suction. Collect the

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precipitate in the crucible, washing with ammonium hydroxide (1:19). Dry the crucible and contents at low heat. Ignite the precipitate to $Mg_2P_2O_7$ by heating at 1000°C to constant weight. Calculate as follows:

$$\text{Percent phosphoric anhydride } (P_2O_5) = 63.774 \frac{W_2}{W_1}$$

Where:

W_1 = Weight in grams of passivator in aliquot.

W_2 = Weight in grams of $Mg_2P_2O_7$.

4.3.4 Insoluble matter. Passivator shall be tested for insoluble matter as follows.

4.3.4.1 Apparatus.

(a) Beaker, 800-mL capacity.
(b) Gooch crucible, porcelain, Coors No. 3, or equal.
(c) Drying oven.

4.3.4.2 Procedure. Weigh 50.0 grams of the passivator to the nearest 0.1 gram into an 800-mL beaker. Add 450 mL of water. Stir and heat the solution to 60°C to dissolve the passivator. Filter the solution through a Gooch crucible with a glass fiber filter disc which has been dried at 110°C and weighed. Transfer all insoluble material to the crucible and wash with 50 mL of water. Dry the crucible and contents for 3 hours at 110°C and weigh. Calculate as follows:

$$\text{Percent insoluble matter} = 100 \frac{W_2}{W_1}$$

Where:

W_1 = Weight in grams of the passivator sample.

W_2 = Weight in grams of the insoluble material.

4.4 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Preservation-packaging and packing. The passivator shall be preserved-packaged and packed level A, B, or C, as specified (see 6.2).

5.1.1 Levels A, B, and C. Under levels A, B, or C, 100 pounds (see 6.2) of passivator shall be preserved-packaged and packed in fiber drums in accordance with PPP-D-723 provided with a 2 mil minimum thickness polyethylene bag type liner as follows:

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Level A - Military overseas drum, type III, grade A.
Level B - Normal overseas drum, type II, grade A.
Level C - Domestic service drum, type I, grade A.

5.2 Palletizing. When specified (see 6.2), drums shall be palletized in accordance with MIL-STD-147.

5.3 Marking. In addition to the special marking specified in 5.3.1 and any other markings specified (see 6.2), the shipping containers shall be marked in accordance with MIL-STD-129. Each shipment shall be identified as to product, assay, net weight, name of manufacturer, date of manufacture and brand name. Each container shall show a lot number of identification of manufacture. In addition to the yellow "OXIDIZING MATERIAL" label specified in section 173.410 of Title 49 of the Code of Federal Regulations, warning labels shall be affixed to interior packages and exterior shipping containers in accordance with 29 CFR, Part 1910.1200.

5.3.1 Special marking. The side of each container shall be durably and legibly marked as follows:

"WARNING
STRONG OXIDANT
FUME AND FIRE HAZARD MIXTURE
CONTACT WITH ORGANIC MATERIAL MAY CAUSE FIRE

Store in dry area. Store separately from oil, grease, gasoline, paint or any combustible material such as wood, paper or cotton. When not in use, keep tightly closed in original container.
Keep away from fire.
Keep away from acids.
Keep away from ammonia and ammonium salts."

5.3.1.1 Disposal instructions. The side of each container shall be durably and legibly marked as follows:

"PASSIVATOR CONTAINS SODIUM NITRITE WHICH CAN BE AN ENVIRONMENTAL POLLUTANT. FOLLOW APPROVED DISPOSAL INSTRUCTIONS WHEN DISCARDING THIS MATERIAL." (Disposal procedures may be found in the Naval Ships' Technical Manual, S9086-T8-STM-000, Chapter 593.)

6. NOTES

6.1 Intended use. The passivator covered by this specification is intended for use as a passivation agent for boiler watersides as the final step of chemical cleaning.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Level of preservation-packaging and packing required (see 5.1).
(c) Quantity required in pounds.
(d) Quantity per container if other than specified (see 5.1.1).
(e) When palletizing is required (see 5.2).
(f) Additional marking if required (see 5.3).

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6.3 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed material safety data sheets prepared in accordance with FED-STD-313 and 29 CFR 1910.1200. The pertinent government mailing addresses for submission of data are listed in appendix B of FED-STD-313.

6.4 Subject term (key word) listing.

Boilers
Disodium phosphate
Monosodium phosphate
Passivator
Phosphoric anhydride
Sodium nitrite

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6850-N794)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-24138A(SH)	PASSIVATOR COMPOUND, NAVY BOILER

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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